

Wiring Beats Blending: What Transfers Between Transformer Sizes — and What Doesn't

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Abstract

Model families train every size from scratch. Can a pretrained large model be converted into a smaller sibling? We characterize the 1.4B→410M conversion in the Pythia family end-to-end: (i) representations align strongly across sizes (ridge $R^2 = 0.84$) while parameters align weakly; (ii) dense weight projection is functionally destructive — provably not an assembly artifact — because basis mixing breaks rotary, per-head, GELU, and Layer-Norm structure; (iii) after the best-fit linear operator, weight residuals are statistically indistinguishable from noise under shuffle controls; (iv) conversion value therefore lives in initialization: in matched-budget continued pre-training we decompose conversion into two independent levers — least-squares compensation (function: best zero-shot) and variance-preserving rescale (training dynamics: best endpoints) — whose combination dominates the strongest subcloning variant on both axes and in every seed (83.9 ± 1.8 vs. 89.7 ± 3.7 final perplexity, 3/3 paired wins; $18\times$ better than from-scratch at 30M tokens), holding out-of-domain, at $2\times$ training context, and on a held-out depth-dominated pair (3 seeds). Code, checkpoints, and the frozen evaluation corpus are released.

1 Introduction

2 Related work

Downscaling pretrained models. The closest prior work is Weight Subcloning (Samragh et al., 2023), which initializes a smaller transformer by ranking units of a larger pretrained one, copying the selected rows and columns, and rescaling — establishing that selection-based initialization accelerates training of the target size. Sheared LLaMA (Xia et al., 2024) learns structured pruning masks jointly with continued pre-training and produces strong small models at a fraction of from-scratch compute; it optimizes the full pipeline, whereas we hold the training recipe fixed across arms and isolate what the initialization alone contributes. Our closed-form compensation belongs to the pruning-with-reconstruction lineage: adjusting surviving weights to absorb the function of removed ones dates to Optimal Brain Surgeon (Hassibi & Stork, 1993) and reappears in structured LLM pruning such as LLM-Pruner (Ma et al., 2023); we apply the reconstruction step once, in closed form, at conversion time. Knowledge distillation (Hinton et al., 2015) transfers function through a teacher’s outputs and requires a full training run; we study the complementary question of what transfers through weights at negligible cost, and the two are combinable.

Growing pretrained models. The mirror direction — initializing a larger model from a smaller one — has a longer history: Net2Net (Chen et al., 2016) introduced function-preserving widening and deepening, bert2BERT (Chen et al., 2022) adapted it to transformers, staged training (Shen et al., 2022) formalized growth operators that preserve loss and training dynamics, and LiGO (Wang et al., 2023) learns a structured linear map from the small model’s weights to the large initialization. That linear operators carry useful signal upward is consistent with our finding that, downward, even a fitted dense projection initializes far better than random; the asymmetry we document is that dense basis mixing destroys the rotary, per-head, and normalization structure that selection preserves (§5).

Representation similarity and parameter symmetries. Our analyses use standard similarity tools: CKA (Kornblith et al., 2019) and SVCCA (Raghu et al., 2017) measure alignment between representations, and model stitching (Lenc & Vedaldi, 2015; Bansal et al., 2021) tests functional interchangeability through a trained adapter. Work on permutation symmetries and model merging (Ainsworth et al., 2023) shows that networks are naturally compared modulo the transformations under which the architecture is invariant. Our selection-versus-blending result is an instance of the same principle: structured selection composes a permutation — an element of the architecture’s symmetry group, restricted to respect head boundaries and rotary frequency pairs (Su et al., 2024) — with coordinate deletion, so every surviving unit keeps the exact nonlinear and positional semantics the architecture assigns it; dense projection mixes coordinates, exits the group, and is punished for it.

Positioning. Each ingredient above exists in isolation: selection-based initialization, least-squares reconstruction, linear growth operators, and representation similarity analysis. What is new here is the setting — conversion between sizes of a single model family trained on identical data, removing data and tokenizer confounds — together with rotary-frequency-matched selection and a controlled end-to-end characterization, from representation alignment through projection diagnosis and residual null tests to a matched-budget training race. Because our subcloning baseline re-implements the recipe of Samragh et al. (2023), we disclose three fidelity differences and label the baseline subcloning-style: (a) we score attention heads and MLP units by weight norms and residual lanes by activation variance, where they score by activation magnitudes; (b) when depth must shrink we remove blocks at even stride rather than from the middle; and (c) we evaluate their $\sqrt{d/d'}$ weight rescale directly rather than adopting it wholesale (§??) — applied to every matrix it collapses zero-shot quality because LayerNorm already renormalizes most read paths, while its variance-preserving motivation holds exactly on the projections no norm protects, the same two families our compensation re-fits.

3 Experimental setup

Model family and conversion pairs. We study the Pythia suite (Biderman et al., 2023): a family of GPT-NeoX models trained on identical data (the Pile) with a shared tokenizer, which isolates size as the only variable between family members. Our primary pair converts 1.4B $\rightarrow$ 410M (24 layers and 16 heads in both; only widths shrink: residual 2048 $\rightarrow$ 1024, head dim 128 $\rightarrow$ 64, MLP 8192 $\rightarrow$ 4096), so layer correspondence is trivially one-to-one and the analysis isolates width. A held-out pair, 410M $\rightarrow$ 160M, exercises the axes the primary pair does not: depth halves (24 $\rightarrow$ 12 blocks), the head count drops (16 $\rightarrow$ 12; head dim unchanged), and width cuts are mild ($\sim 25\%$). A third pair, 6.9B $\rightarrow$ 1.4B, tests scale (all three axes reduced; §??).

Frozen evaluation corpus. All representation analyses probe every model with the same inputs: 10,000 sequences of 128 tokens drawn from the Pile (pile-uncopyrighted), tokenized once and frozen for the entire project. The first 500 sequences (all token positions) form the subset for similarity analysis (§4); the remainder contribute mean-pooled and last-token activations. Continued pre-training streams the same corpus but skips the region used to build the frozen set (a contamination guard), and each random seed trains on a disjoint stream offset.

Behavioral evaluation. Language-modeling quality is strided sliding-window perplexity on WikiText-103 validation (every token scored exactly once). Hardening evaluations add C4 perplexity (out-of-domain), perplexity at $2\times$ the training context (2048), and zero-shot accuracy on LAMBADA, ARC-Easy, HellaSwag, and PIQA via lm-eval. Reference models anchor every table; anchor scores match published Pythia numbers.

Determinism and verification. All forward passes are greedy and seeded; repeated runs are bit-identical on the same device. Weight extraction is validated by exact reconstruction: the fused GPT-NeoX QKV matrices are split head-aware and re-fused to bit-identical equality, and a model rebuilt from its own extracted weights reproduces the original logits with zero

difference (§5) — any quality change downstream is therefore attributable to conversion mathematics, not assembly.

Training protocol. Every conversion arm within an experiment shares identical data order, optimizer (AdamW, $\beta = (0.9, 0.95)$, weight decay 0.1), cosine schedule with 100-step warmup, gradient clipping at 1.0, bf16 autocast, and token budget (30M primary; 100M persistence checks); the learning rate equals the target size’s original pretraining rate. Initialization construction costs are negligible against any training budget (selection: seconds; moments pass for compensation: ~ 75 s; closed-form solves: seconds) and are included in the compute accounting. All experiments ran on a single NVIDIA GB10 (128 GB unified memory); any 24 GB+ CUDA GPU reproduces them (Appendix TODO).

4 Representations align, parameters do not

A size conversion presupposes that the two family members are related; the question is where the relation lives. On the primary width-only pair (1.4B $\rightarrow$ 410M, same depth and head count; §3) we measure the relation in two places — the representations both models compute, and the parameters they store — and find it in only one of them.

Layer correspondence saturates under CKA. Linear CKA between all 24×24 layer pairs, computed on the similarity subset (all token positions; §3), gives a diagonal mean of 0.883, yet only 12% of layers place their single best match exactly on the diagonal. The reason is visible in Figure 1 (left): the heatmap is one large saturated block — layers 0–2 distinct, the middle band (layers ~ 4 –22) mutually ≈ 1.0 , layer 23 distinct. Linear CKA therefore confirms only coarse early/mid/late correspondence and saturates in the middle band (a known limitation of the metric), so it is not a sharp per-layer discriminator. This motivates a directional, quantitative map: how much of the small model’s representation is a linear image of the large model’s?

Activations are largely a linear image across widths. Per layer, we fit a large $\rightarrow$ small map on pooled activations (8k train / 2k held-out) and score held-out R^2 under two operators of increasing freedom: Procrustes (a rotation plus one global scale) and ridge (a full linear map $2048 \rightarrow 1024$). Ridge reaches a mean held-out $R^2 = 0.844$ (Table 1): the 410M pooled representation is largely a linear projection of the 1.4B’s, capturing 84% of held-out variance. Procrustes already reaches 0.717, so most of the relationship is a change of basis; the ~ 0.13 gap is genuine non-orthogonal reshaping. Both maps are discriminative per layer and dip together at layers 3–5 — the same layers CKA flagged as distinct (Figure 1, right).

Raw parameters do not. The token embeddings are the one weight matrix where the two sizes share an axis (the vocabulary) and differ only in width, so a single map applies; we split rows (tokens) 80/20. Even in this cleanest case a linear map explains only 0.25–0.39 of weight variance (Table 1): ridge $R^2 = 0.248$ on the input embedding and 0.386 on the output embedding, against 0.844 for activations.

The representation–parameter gap. Two independently trained members of a family arrive at closely related representations while their parameters are only loosely a linear map of one another. This gap is the pivot of the paper. If representations align but weights do not, then projecting the large model’s weights into the small model’s shape (§5) must leave a substantial residual, and whether that residual is structured and learnable or merely noise (§6) decides where conversion value can come from.

5 Why dense projection fails

If representations align across sizes, the natural attempt is to project the large model’s weights into the small model’s shape and read off a working small model. We formulate this projection using only the size relationship (never the target’s weights), measure the residual, run the assembled model, and diagnose why it breaks. The verdict is that dense

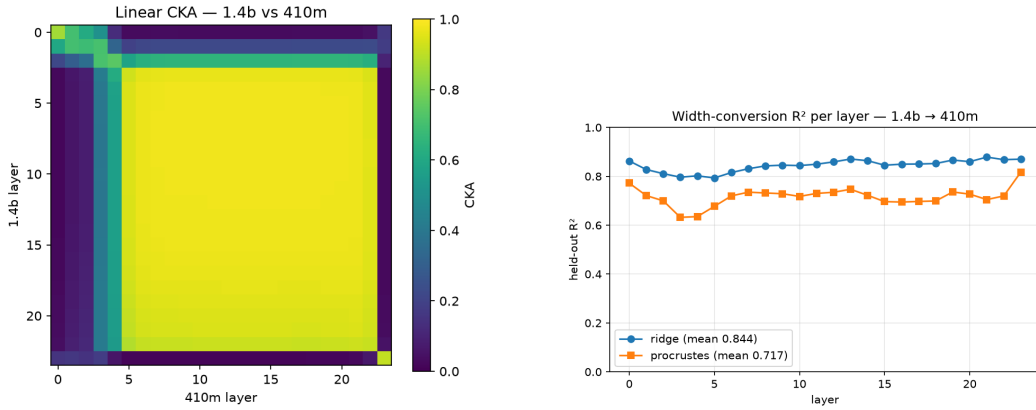


Figure 1: Alignment across sizes for the primary pair (1.4B $\rightarrow$ 410M). (left) Linear CKA between all 24×24 layer pairs: a saturated middle band (layers ~ 4 –22 mutually ≈ 1.0) confirms coarse correspondence but resolves no sharp per-layer match. (right) Held-out R^2 of the per-layer activation maps: ridge (full linear) and Procrustes (rotation+scale) track each other and dip together at layers 3–5, the layers CKA also flags as distinct.

Table 1: Alignment across sizes, held-out. Activations align strongly under a linear map; raw parameters, even in the cleanest (embedding) case, do not. Best (highest) map per row in bold.

Target	CKA	Ridge R^2	Procrustes R^2
Pooled activations (per-layer mean)	0.883 [†]	0.844	0.717
Input embedding	0.512	0.248	0.167
Output embedding (unembedding)	0.678	0.386	0.242

[†] For activations, the diagonal mean of the 24×24 linear-CKA matrix, which saturates in the middle band (Figure 1, left); for embeddings, linear CKA before mapping.

blending destroys the network’s structured computation — and, via a bit-exact control, that the destruction is in the projection mathematics, not the assembly.

Target-free projection. Each weight is mapped by a pair of structured linear operators,

$$\hat{W} = P_{\text{out}} W P_{\text{in}}^{\top}, \quad P_{\bullet} \in \{P_{\text{res}}, P_{QK}, P_{VO}, P_{\text{MLP}}\}, \quad (1)$$

whose bases are built only from the large model and the size relationship, never from the 410M target. The residual-stream axis uses a global activation-PCA basis P_{res} ; the internal axes use SVD bases shared between interacting weights, so that composed operations stay consistent — Q and K share one basis, V and O share one, and MLP_UP/MLP_DOWN share a single joint basis computed from the column stack $[\text{UP}, \text{DOWN}^{\top}]$. Sharing is not cosmetic: a per-weight basis for the MLP hidden axis cannot span the target, since one 8192×2048 matrix has rank $\leq 2048 < 4096$, so the interacting pair must be factorized together.

The projected weights miss the target solution. Table 2 reports the relative Frobenius error of $\hat{W}$ against the actual 410M weights, averaged over 24 layers, for the target-free projection of Eq. 1 and — as a diagnostic upper bound — a shared operator (A, B) fitted with access to the target. Every target-free error exceeds 1, i.e. worse than predicting zero: the projected weights share almost nothing with the 410M’s particular solution, confirming the activation-versus-weight gap of §4 at every layer. Even the fitted operator, which peeks at the target, only reaches 0.66–0.80; at best $\sim 30\%$ of weight variance is linearly explainable across sizes with one shared map.

Table 2: Relative Frobenius error of projected weights vs. the actual 410M weights, mean over 24 layers. Every target-free error exceeds 1 (worse than predicting zero). Lower error per row in bold.

Weight	Projection (target-free)	Operator (fitted) [†]
Q	1.471	0.658
K	1.591	0.758
V	1.390	0.804
O	1.724	0.795
MLP_UP	1.668	0.683
MLP_DOWN	1.723	0.686

[†] A single (A, B) shared across layers per type; it peeks at the target and so upper-bounds the linearly explainable fraction. A per-layer fit is degenerate (error $\rightarrow 0$ around any full-rank W) and is not reported.

Table 3: Zero-shot language-modeling quality of the dense-projected model (strided WikiText-103 perplexity). The projected model does not beat random init. Best (lowest) in bold.

Model	Perplexity
real 1.4B (reference)	11.5
real 410M (upper anchor)	15.6
random-init 410M (lower anchor)	65,962
projected, LN copied from 410M	181,242
projected, LN projected from 1.4B	10^{13}

With fresh neutral LayerNorms (gain 1, bias 0) the same projected matrices score $\sim 12.3k$ quick-perplexity — $\sim 15\times$ better than the 181,242 row and $\sim 5\times$ better than random init — so much of the collapse is LayerNorm mismatch, not weight destruction alone (§??); the structure-mixing diagnosis still holds directionally.

Behavioral anchors. We assemble a 410M-shaped model from the projected 1.4B weights and score strided WikiText-103 perplexity against reference and anchor models (Table 3). Greedy generation is deterministic but word salad, and the acceptance bar — beat random init — is not met zero-shot. The shape of the failure is instructive: random init produces near-uniform logits (perplexity on the order of the vocabulary size), whereas the projected model is confidently wrong, which scores far worse than uniform.

The failure is projection math, not assembly. Before diagnosing the mechanism, we rule out a plumbing bug. Rebuilding the 1.4B from its own extracted weights with no projection reproduces the real model’s logits with $\max|\text{diff}| = 0$ — bit-exact (previewed in §3). Extraction, head-aware QKV split/fuse, embeddings, and LayerNorm plumbing are all correct, so the entire quality loss in Table 3 is attributable to Eq. 1.

Structure mixing breaks the computation. A dense basis change on any internal axis silently changes what the block computes, because every structured or nonlinear operation is tied to specific coordinates. Rotary embeddings act on fixed per-head dimension pairs, so any head-space mixing distorts positions. Attention is per-head, yet our global $2048 \rightarrow 1024$ head-space basis mixes dimensions across heads. GELU is elementwise in the MLP hidden space, so $\text{GELU}(Bx) \neq B \text{GELU}(x)$. LayerNorm gains and biases are per-feature in the residual stream, so a rotated stream no longer matches any diagonal gain — which is why the projected-LN variant explodes to 10^{13} . The consequence is that function-preserving zero-shot conversion needs structure-respecting selection maps — keep or drop whole heads, hidden units, and features, the architecture’s own symmetry group — rather than dense blending. Dense learned projections are thus useful as initializations for brief fine-tuning, not as drop-in models; §?? takes up exactly that comparison.

6 Post-operator residuals are noise

Section 5 shows that even the strongest shared linear operator explains at best $\sim 30\%$ of weight variance (relative error 0.66–0.80), leaving the majority as a residual. Is that residual a structured, learnable correction — or noise? We test it against matched controls, under the strongest-alignment condition so the verdict is conservative: if any signal survives the best linear map, these tests should find it.

The residual under the strongest operator. We define the residual per weight as

$$\Delta(l, \text{type}) = W_{410M} - A W_{1.4B} B^\top, \quad (2)$$

using the shared fitted operator (A, B) per type — the strongest-alignment condition of §5, so whatever remains is what no single linear size-map can explain. Four tests probe Δ , each against a matched control.

Test 1 — spectral (effective rank). Table 4 compares the effective rank of Δ against a per-type shuffled control and a shape/scale-matched Gaussian control. Δ sits 3–6% below both controls in every type: a faint spectral concentration, so the residual is not perfectly isotropic noise (Figure 2).

Test 2 — cross-layer consistency. The mean pairwise cosine of vectorized Δ across layers is $\approx \pm 0.0002$ in every type, identical to the shuffle control: no direction is shared across layers, so there is no single correction the projection is uniformly missing.

Test 3 — learnability (decisive). A patch predictor $R(\hat{W} \text{ patch} + \text{type/layer/position}) \rightarrow \Delta \text{ patch}$ is trained on 18 layers and scored on 6 stratified held-out layers as error reduction over predicting zero; the control retraines identically on Δ shuffled across layers, and a 1000-sample bootstrap brackets the real–control gap. Neither a linear nor an MLP-512 predictor beats predicting zero (reductions $-0.0004/-0.0003$), and real and control are indistinguishable (gap $\text{CI}_{95} \approx [-0.0000, -0.0000]$, $p \geq 0.99$; Table 5). There is no patch-learnable signal.

Test 4 — transfer. On the held-out pair 410M $\rightarrow$ 160M the shared operators fit at relative error 0.56–0.74 (better than the primary pair — a smaller width gap). A predictor trained on all primary-pair layers and evaluated on the held-out pair reduces error by $-0.0004/-0.0003$, again indistinguishable from its shuffle control ($p = 1.00$; Table 5). Nothing transfers, consistent with Test 3.

Reconciling the faint spectral signal. Tests 1 and 3 are not in tension. The 3–6% spectral deficit is a generic statistical trace — mild row/column-norm heterogeneity, for instance — not a correspondence between $\hat{W}$ and Δ that any predictor can exploit. A behavioral cross-check agrees: an assembled 410M whose six matrix types are operator-projected (with real biases, LayerNorms, and embeddings) scores 35,212 perplexity, and the learned correction moves it only $\sim 2\%$ (to 34,373); both remain non-functional. Notably the operator matrices alone beat random init (35k vs. 66k) once the surrounding tensors are real — unlike §5’s fully projected model — which again locates the destruction in the dense projection, not the residual.¹

Handoff to §??. Within a model family the sizes share representation geometry but not parameter solutions, and what the best linear size-map misses behaves as noise: no cross-layer direction, no patch-learnable mapping, no cross-pair transfer, only a faint generic spectral fingerprint. A learned zero-shot correction is therefore not available at this granularity. Conversion value must instead come from initialization plus brief fine-tuning — the matched-budget recovery race of §??.¹

¹The correction’s gains on the 18 training layers are partly memorized; only the held-out layers carry genuine predictions, and there the reduction is zero.

Table 4: Test 1 (spectral). Effective rank of the residual Δ vs. a per-type shuffled control and a shape/scale-matched Gaussian control. Δ sits 3–6% below both in every type — a faint, non-isotropic concentration, not a learnable direction.

Weight	Δ erank	Shuffled	Gaussian
Q	771.3	824.1	824.1
K	784.0	824.0	824.1
V	801.2	824.2	824.1
O	792.2	824.1	824.1
MLP_UP	947.2	989.4	989.4
MLP_DOWN	934.8	989.4	989.4

Table 5: Tests 3–4 (learnability and transfer). Patch-predictor error reduction over predicting zero, vs. a label-shuffled control; 1000-sample bootstrap CI on the real–control gap. No predictor beats zero, and none separates from its control.

Setting	Predictor	Reduction	Shuffle	Gap CI ₉₅	p
Held-out layers (1.4B→410M)	linear	−0.0004	−0.0004	[−0.0000, −0.0000]	0.99
	MLP-512	−0.0003	−0.0003	[−0.0000, −0.0000]	1.00
Transfer (410M→160M)	linear	−0.0004	−0.0004	—	1.00
	MLP-512	−0.0003	−0.0002	—	1.00

CIs are reported for the in-pair held-out test only; the transfer test reports p (M4). Reductions are negative because no predictor improves on predicting zero.

7 Matched-budget conversion: the two levers

8 Limitations and future work

Reproducibility statement

References

- Samuel K. Ainsworth, Jonathan Hayase, and Siddhartha Srinivasa. Git Re-Basin: Merging models modulo permutation symmetries. In International Conference on Learning Representations (ICLR), 2023.
- Yamini Bansal, Preetum Nakkiran, and Boaz Barak. Revisiting model stitching to compare neural representations. In Advances in Neural Information Processing Systems 34 (NeurIPS), 2021.
- Stella Biderman, Hailey Schoelkopf, Quentin Anthony, Herbie Bradley, Kyle O’Brien, Eric Hallahan, Mohammad Aflah Khan, Shivanshu Purohit, USVSN Sai Prashanth, Edward Raff, Aviya Skowron, Lintang Sutawika, and Oskar van der Wal. Pythia: A suite for analyzing large language models across training and scaling. In Proceedings of the 40th International Conference on Machine Learning (ICML), pp. 2397–2430. PMLR, 2023.
- Cheng Chen, Yichun Yin, Lifeng Shang, Xin Jiang, Yujia Qin, Fengyu Wang, Zhi Wang, Xiao Chen, Zhiyuan Liu, and Qun Liu. bert2BERT: Towards reusable pretrained language models. In Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL), 2022.
- Tianqi Chen, Ian Goodfellow, and Jonathon Shlens. Net2Net: Accelerating learning via knowledge transfer. In International Conference on Learning Representations (ICLR), 2016.
- Babak Hassibi and David G. Stork. Second order derivatives for network pruning: Optimal brain surgeon. In Advances in Neural Information Processing Systems 5 (NIPS), pp. 164–171, 1993.

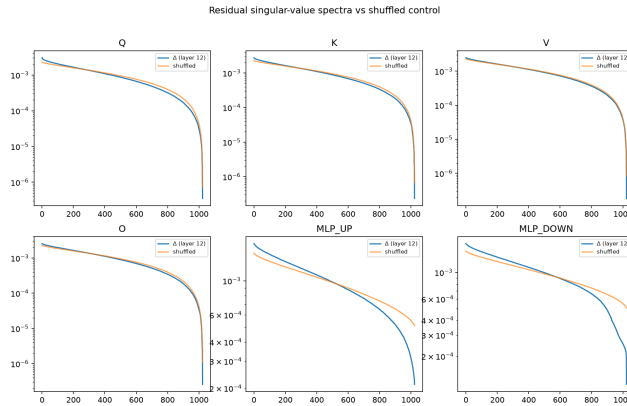


Figure 2: Test 1 (spectral). Singular-value spectrum of the residual Δ against shuffled and Gaussian controls. Δ concentrates 3–6% more than either control in every weight type — a faint, generic non-isotropy that carries no layer-specific, predictable signal (Table 5).

Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the knowledge in a neural network. arXiv preprint arXiv:1503.02531, 2015.

Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. Similarity of neural network representations revisited. In Proceedings of the 36th International Conference on Machine Learning (ICML), pp. 3519–3529. PMLR, 2019.

Karel Lenc and Andrea Vedaldi. Understanding image representations by measuring their equivariance and equivalence. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp. 991–999, 2015.

Xinyin Ma, Gongfan Fang, and Xinchao Wang. LLM-Pruner: On the structural pruning of large language models. In Advances in Neural Information Processing Systems 36 (NeurIPS), 2023.

Maithra Raghu, Justin Gilmer, Jason Yosinski, and Jascha Sohl-Dickstein. SVCCA: Singular vector canonical correlation analysis for deep learning dynamics and interpretability. In Advances in Neural Information Processing Systems 30 (NeurIPS), 2017.

Mohammad Samragh, Mehrdad Farajtabar, Sachin Mehta, Raviteja Vemulapalli, Far-tash Faghri, Devang Naik, Oncel Tuzel, and Mohammad Rastegari. Weight sub-cloning: direct initialization of transformers using larger pretrained ones. arXiv preprint arXiv:2312.09299, 2023.

Sheng Shen, Pete Walsh, Kurt Keutzer, Jesse Dodge, Matthew Peters, and Iz Beltagy. Staged training for transformer language models. In Proceedings of the 39th International Conference on Machine Learning (ICML), pp. 19893–19908. PMLR, 2022.

Jianlin Su, Murtadha Ahmed, Yu Lu, Shengfeng Pan, Wen Bo, and Yunfeng Liu. RoFormer: Enhanced transformer with rotary position embedding. Neurocomputing, 568:127063, 2024.

Peihao Wang, Rameswar Panda, Lucas Torroba Hennigen, Philip Greengard, Leonid Karlinsky, Rogerio Feris, David D. Cox, Zhangyang Wang, and Yoon Kim. Learning to grow pretrained models for efficient transformer training. In International Conference on Learning Representations (ICLR), 2023.

Mengzhou Xia, Tianyu Gao, Zhiyuan Zeng, and Danqi Chen. Sheared LLaMA: Accelerating language model pre-training via structured pruning. In International Conference on Learning Representations (ICLR), 2024.